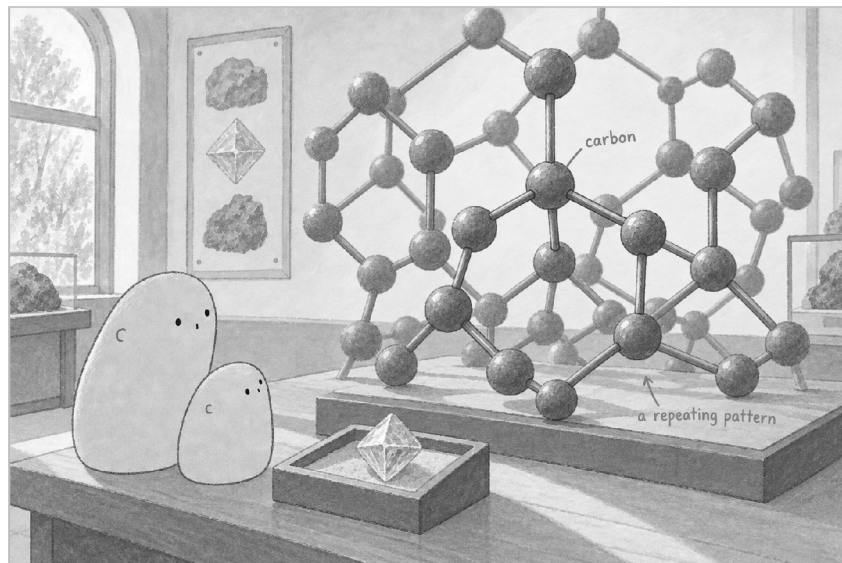


Parts for a pattern

The flash that wouldn't stay. · Curiosity-led family day

Name _____

Date _____



The model reveals a repeating carbon pattern—not a hidden source of light. Real atoms are much too small to see.

PRETEND MODEL SUPPLIES · RODS

Tray	A	B
Rods	18	7

Use these invented supply counts. Do not count the rods in the painting.

1. Tray A has 18 rods. Tray B has 7. How many rods altogether?

$$18 + 7 = \underline{\quad}$$

Draw or write your thinking.

2. You have 25 rods. Set 9 aside for later. How many can you use now?

$$25 - 9 = \underline{\quad}$$

Draw or write your thinking.

3. Draw 3 separate groups with 4 rods in each. How many rods did you draw?

$$4 + 4 + 4 = \underline{\quad}$$

Draw or write your thinking.

These are invented supply counts, not a count of atoms in a diamond. Try drawing a different way to group 12 rods.

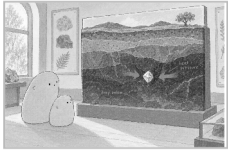
GROWN-UP NOTE About 15 min: 3 to explore the model and table, 9 to draw or calculate, 3 to explain. Use pencil marks or number-line hops; no loose model pieces needed. Accept a spoken strategy. Mohs ranks are not used as amounts in these sums.

The flash that wouldn't stay

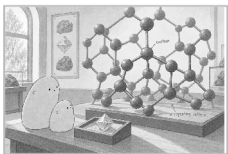
The flash that wouldn't stay. • Curiosity-led family day

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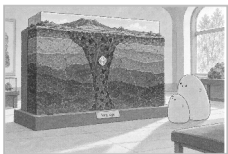
Date _____



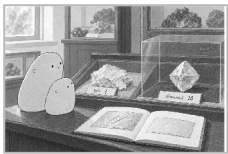
A flash winks from a museum case. "The diamond makes light!" says Little Wonderer. Big points to its deep beginning. Heat and huge pressure helped carbon grow into a crystal far underground—without lamp or sun. "Then where does the flash come from?"



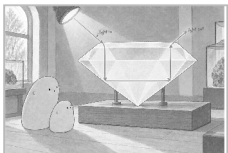
"Maybe light hides in the tiny parts," Little says. Balls and rods stand for carbon atoms and their connections. The pattern repeats through a diamond—but it is no lantern. "We found the stone," says Big, "not the flash."



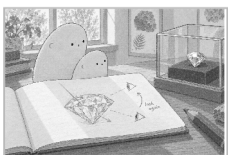
Little traces an arrow up. "To meet light, it had to leave the dark!" Long after some diamonds formed, volcanic eruptions carried them toward the surface. "Now it can meet light," says Little. Big nods. "But meeting light is not making it."



A label marked 10 stops Little. Diamond tops the Mohs scratch scale. "So nothing can hurt it?" Big explains that a diamond can still chip or break. "Hard to scratch is not unbreakable," Little says, "and it still doesn't explain the flash."



At the last model, a lamp clicks on. "There!" Little cries. Light enters the diamond, bends, bounces, and comes out again. Cut, flat faces called facets guide its path. Little follows the line. "The diamond sends borrowed light back!"



Little shifts. One bright face dims; another flashes. "Found you!" The flash was not stored in the stone. It happens when light, cut facets, and watching eyes meet along the right paths. Little sketches both spots, leaving room for a new question.

Talk together: What did Little first think made the flash? Which clue changed that idea? At the end, what had to meet for Little to see a flash?

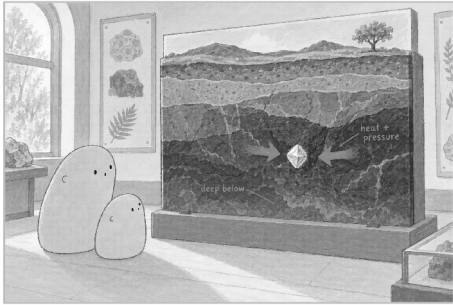
Keep a clue from the chase.

The flash that wouldn't stay. • Curiosity-led family day

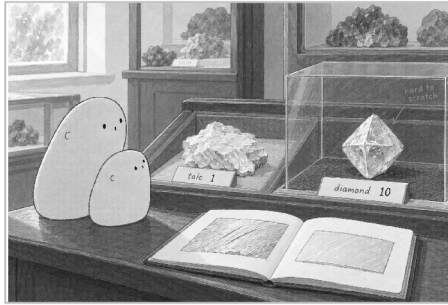
Name _____

Date _____

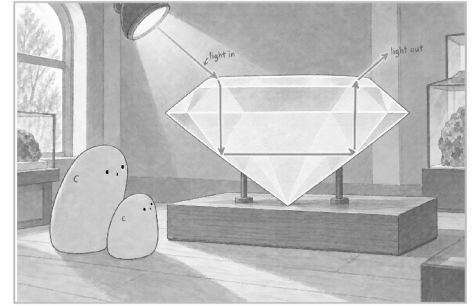
Choose a picture from Little's search for the flash. Say what changed Little's mind. Then draw or write one idea you want to keep.



No sun down here



The strong-stone surprise



Where the flash begins

Draw one detail from your scene. Add an arrow or a label to help someone understand it.

carbon • crystal • heat • pressure • scratch • light • facet

Deep underground, ... What helped the diamond crystal grow?

A diamond is hard to scratch, but ... What could still happen to it?

The flash begins when ... What gives the light? How do the facets and watching eyes help?

Read it together. Point to the detail your words explain. If you are writing, check the first capital and the ending mark.

GROWN-UP NOTE About 15 min: 3 to choose and talk, 4 to draw, 6 for 1–3 sentences, 2 to read back. One careful idea is enough. Accept invented spelling, speech, or dictation. Help Leo keep his own phrasing.